3.13 Let A, B, C be subsets of a set X. Which of the expressions $(A \cap B) \cup (A \cap B^c)$, $(A \cap B) \cup (A \cap B^c)^c$, $(A \cap B) \cup (A^c \cup C)^c$ can be simplified, and how?

3.4.2 Let g be the mapping from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined by

$$g(x, y) = (y, -x),$$

and let the mapping h be as in Exercise 3.4.1. Find the image of (x, y) under each of the composite mappings $g \circ h$ and $h \circ g$.

31.1.1 As in the text, let $\mathbb{Q}$ be the set of all rational numbers. If $x \in \mathbb{Q}$, which of the following statements (P, Q, R, S) imply which of the others?

P: $x \geq 0$

Q: $\exists w \in \mathbb{Q}$ such that $x = w^2$

R: $nx > -1 \ \forall n \in \mathbb{N}$

S: $\exists n \in N$ such that $nx > -1$.

5.1.3 In each of the following cases, state whether the sequence $\{u_n\}$ tends to a limit, and find the limit if it exists:

(a) $u_n = 1 + \frac{1}{2}n$

(b) $u_n = 1 - \frac{1}{2}n$

(c) $u_n = (\frac{1}{2})^n$

(d) $u_n = -(\frac{1}{2})^n$